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Robustness of Venture-Capital Exit Prediction to Structural Perturbations in Investor Co-Investment Networks: A Comparison of Tabular and Graph Neural Network Models

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Robustness of Venture-Capital Exit Prediction to Structural Perturbations in Investor Co-Investment Networks: A Comparison of Tabular and Graph Neural Network Models

by

Karim Abbas

September 2026

Dissertation Supervisor: Li Zhang

Dissertation submitted in part fulfilment of the

Degree of Master of Science in Venture Capital, Private Equity with

Financial Technology

Department of Finance and Technology

University College London

DECLARATION

I, Karim Abbas, hereby declare that the work presented in this dissertation is my own original work. Where information has been derived from other sources, I confirm that this has been clearly and fully identified and acknowledged. No part of this dissertation contains material previously submitted to the examiners of this or any other university, or any material previously submitted for any other assessment.

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ABSTRACT

Applications of graph neural networks (GNNs) for venture capital exit prediction are growing, as these models can capture relational information that tabular features cannot represent. Most of these models, however, are built on a single fixed snapshot of the network. This fails to capture how VC co-investment ties actually form and dissolve over time. Prior research in this space has not tested how stable predictions are. This study tests the robustness of VC exit predictions to structural perturbations, using Refinitiv US VC deal data from 2000 to 2024 and a classification sample of 11,791 startups drawn from the 2010–2017 first-round cohort. It tests whether network-based features and a GNN improve predictive performance over standard tabular models. Across all eleven model specifications, ROC-AUC ranged only from 0.64 to 0.66, and none of the differences between models reached statistical significance. Models were mildly affected by random perturbations to the network but far more severely affected by targeted removal of highly connected investors. The GNN's ROC-AUC stayed stable across most perturbation types, but its individual predictions were twice as likely to flip as the average tabular model's. The central contribution of this study is therefore not that network features improve predictive accuracy; rather it is that they introduce a distinct kind of instability. Removing a small number of central investors leaves aggregate ROC-AUC nearly unchanged while reshuffling which specific startups the model ranks as likely to exit.

GitHub repository: Venture Capital Co-Investment Structures Codebase

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1. INTRODUCTION

Venture capital is a specialized form of financing dedicated to early-stage, high-growth firms, supporting innovation and driving regional economic growth. The US VC market is the largest globally, accounting for 57% of global venture deal value in 2024, worth $215.4 billion (Ceppos, 2025). The US VC market is highly mature, with $1.25 trillion in AUM, and $207.8 billion in dry powder as of 2024 (NVCA, 2025). Strong capital flows and a large investor base make the US a good setting for empirical analysis.

Venture capital syndication is when multiple investors jointly finance the same firm, creating a co-investment tie. Syndication networks shape capital flows within the venture capital ecosystem (Sorenson and Stuart, 2001). Better-networked VC firms and their portfolio companies have been shown to outperform less-connected counterparts. This is because syndication provides better financing opportunities, monitoring, signaling, and information-sharing functions (Hochberg et al., 2007).

Venture capital firms realize gains when a company successfully exits, providing liquidity to the fund and its limited partners. Exits are typically via IPOs, trade sales, or liquidation events. The exit market in the United States is highly developed, with 2024 exits totaling $98 billion across 1,147 deals. The literature emphasizes that exit timing and type are observable from a firm's characteristics, deal structures, and current market conditions (Giot and Schwienbacher, 2007). VC investing is highly relational. Investors back young firms based on trust and reputation, compensating for limited information, and VC networks shape deal access, signaling, and bargaining dynamics among investors (Hochberg et al., 2007).

Current methodologies for predicting venture capital exits rely on tabular features such as age, funding history, sector, geography, and investor-reputation proxies. These proxies reduce investors to simple measures like "reputation" or "past backing," compressing relational information. This leaves out the network's impact on exit outcomes, including the monitoring, access to follow-on capital, and syndication-based signals carried by repeated co-investment relationships (Kaplan and Lerner, 2016). This means even strong classifiers may learn from an incomplete or inconsistent record rather than the true underlying market structure (Kaplan and Lerner, 2016).

Graph neural networks (GNNs) are designed for data whose outcomes depend on

both entity-level features and the links between entities, making them useful for observing relational structures within financial systems (Wang et al., 2022). VC markets fit this setting because startups and investors are linked through investment relationships (Kojaku and Miyakawa, 2017). Startup outcomes may depend not just on a startup's own attributes, but on the quality and centrality of the investors backing it (Kojaku and Miyakawa, 2017). GNNs can learn from direct and indirect connections across multiple co-investment paths, treating that structure itself as part of the predictive signal rather than discarding it (Wang et al., 2022).

Although GNNs better represent VC networks, graph-based predictions are often built from a fixed network snapshot, while real VC networks evolve over time (Feng et al., 2026). VC co-investment networks are dynamic. Investors and startups constantly enter and exit, with ties dissolving and forming as more firms move across stages. This is especially relevant in VC, as datasets often miss syndicate members, misclassify investors, or record ties differently over time. Commercial VC databases are known to have data gaps, with investor stakes in startups often going unrecorded or a syndicate member being missing from deal records.

Perturbing the GNN can simulate the realistic market conditions VCs and startups face, where co-investment relationships constantly change and are never perfectly observed. GNN predictions are known to be sensitive to small changes in edge or node structure, meaning strong predictive accuracy does not guarantee stable predictions (Xu et al., 2021; Freitas et al., 2023). That sensitivity is why robustness testing matters if these models are to be reliable for predicting startup exits in practice. The literature reviewed here spans VC syndication and network effects on performance, machine learning approaches to exit prediction, and graph-based robustness testing.

This empirical analysis constructs a time-respecting VC co-investment network, compares network features with a heterogeneous GNN, and conducts a structured robustness analysis of both model classes. From the literature reviewed, no study specifically evaluates the stability of VC exit prediction under perturbation, and the adversarial robustness literature has largely been tested on synthetic graphs (Zügner et al., 2018; Lyu et al., 2021; Zhang et al., 2021; Trappolini et al., 2024). By testing whether robustness depends on perturbation type in a real VC co-investment network, this study shows that robustness checks are necessary to validate VC exit-prediction

models before deployment. The empirical design uses a baseline tabular model, then the same model with a network feature added, then a heterogeneous GNN and finally runs perturbation tests. The perturbation tests in this study are designed to check whether exit predictions built on this kind of network are able to survive data incompleteness.

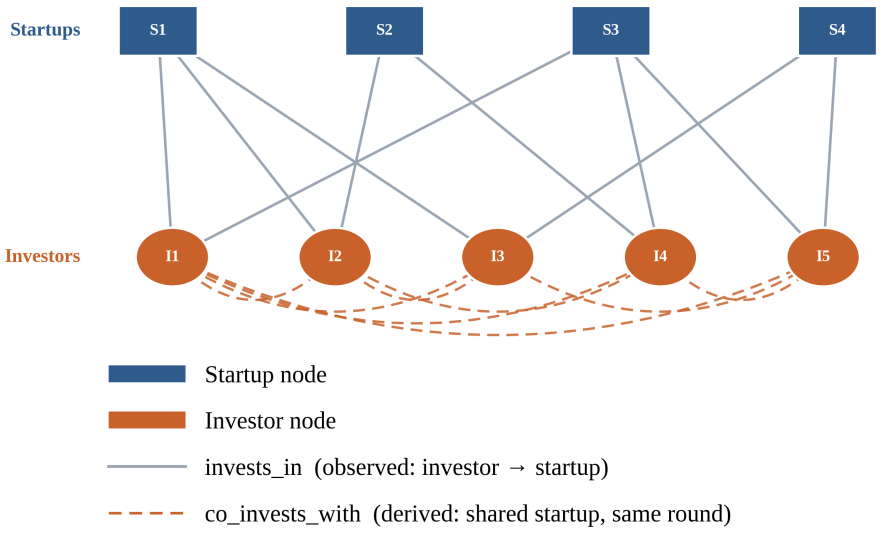


Figure 1.1: Investor-startup network, showing bipartite investment ties and the co-investment layer

2. RESEARCH QUESTIONS AND HYPOTHESES

2.1 Research Question

To what extent do out-of-sample exit predictions for US venture-capital-backed startups depend on investor network structure, and how fragile is that dependence to targeted versus random perturbations of the co-investment and investor–startup networks?

Robustness refers to how stable a model's predictive performance and individual predictions are when the co-investment graph is perturbed via controlled, budgeted changes. The literature indicates that adding network features has mixed effects on the value of VC exit predictions (Section 3.6); hence, the research question is broken down into 3 key hypotheses.

2.2 Hypothesis 1: Network Position Adds Predictive Value

Investor co-investment network centrality (mean PageRank of a startup's first-round investors) provides incremental predictive power for 7-year exit outcomes beyond deal characteristics (first round raise, cohort year, firm age, and sector) alone.

Literature reviewed highlights that better-networked investors are associated with better fund and portfolio-company performance (Hochberg et al., 2007). Centrality in either an investor network (Esposito et al., 2022) or a startup mobility network (Bonaventura et al., 2020) has been shown to carry information that firm-level attributes do not capture. Hypothesis 1 is tested by comparing a baseline tabular model against a network-augmented version of the same model class and checking whether the change in discriminative performance (ROC-AUC) is statistically distinguishable from zero.

2.3 Hypothesis 2: Learned Versus Engineered Network Representation

A heterogeneous graph neural network that learns its own representation of the heterogeneous investor-startup graph will outperform a tabular model with a centrality statistic.

Hypothesis 2 tests whether the way network position is represented matters. A single centrality statistic condenses each investor's position into one number, discarding information from the rest of the graph. Message-passing architectures like GraphSAGE (Hamilton et al., 2017) instead learn representations directly from a node's nearby and farther-out connections. This is consistent with prior applications of heterogeneous graph learning to startup data, where richer relational representations improved predictive performance over feature-based baselines (Zhang et al., 2021; Zhang et al., 2024; Lyu et al., 2025). Hypothesis 2 compares the clean-graph test performance of the heterogeneous GNN against both the baseline and network-augmented tabular models.

2.4 Hypothesis 3: Robustness Depends on Perturbation Types

The network-augmented tabular models and heterogeneous GNNs operating on the graph will be more sensitive to targeted, structural perturbations, such as hub removal, than to

random, low-magnitude perturbations, such as missing or spurious edges.

This follows from the adversarial-robustness literature reviewed in Section 3.5, which shows that sparse, targeted attacks degrade GNN predictions disproportionately relative to random noise of similar scale (Trappolini et al., 2024; Zügner et al., 2018). VC co-investment networks are a relevant test case because they are sparse and hub-dominated: losing a small number of highly connected investors is a different kind of shock than losing an equivalent number of random edges or non-hub investors. To keep that comparison fair, structural perturbation budgets bound how much a model's output can change for a fixed number of edge or node changes (Zügner and Günnemann, 2019b). Hypothesis 3 is evaluated using this perturbation protocol and a set of stability metrics.

3. LITERATURE REVIEW

3.1 Venture Capital, Syndication, and Performance

Economic Role of VC and Exits

The core objective of venture capital funds is to achieve successful exits, which may take many forms, such as an IPO, trade sale, or secondary buyout. Exit likelihood, however, depends on many difficult-to-measure factors, especially at earlier stages. Sorensen (2007) found that more experienced investors are twice as likely to have an eventual IPO. That correlation cannot rule out selection, as a more experienced investor may simply screen better deals to begin with. Sorensen's two-sided matching model separates sorting from influence and attributes most of the differential to sorting. If network position is really just proxying for deal quality, then incremental predictive power tells us the features are informative, not that it is causal. Hochberg et al. (2007) showed that within VC-VC syndication networks, better-networked funds are associated with better fund performance. Better-networked funds might simply have been good funds already, not funds the network improved. The five-year trailing window helps but does not fully solve this, since unobserved fund quality could still explain both the networking and the performance. Bubna et al. (2020) similarly found that venture capitalists cluster in tightly connected communities that shape access to deals and information. None of these three studies, though, tests whether network position adds anything once firm and deal features are already in the model.

Syndication and co-investment networks

Syndication structure influences exits and returns (Hochberg et al., 2007; Bellavitis et al., 2020). Bellavitis et al. (2020) modeled non-linear effects, finding that investors who have previously co-invested together exhibit coordinated behavior and share information more effectively. But the same paper finds that repeated co-investment by the same investors reduces performance, the opposite of what stronger information flow would predict. Some evidence points to VC-syndicate-backed targets receiving higher acquisition premiums and fund returns during M&A exits (Nguyen and Vu, 2021). Tian (2012) finds a similar value-creation result that syndicated VC investment increases a portfolio company's innovation output and patenting relative to a solo-backed firm.

Christopoulos et al. (2022) found that syndicate-backed firms sitting centrally in the network survive longer, supporting the idea that network position matters. But survival and successful exit are not the same thing; a firm that persists for years without ever returning capital counts as a survival success and an exit failure. The finding hence really only supports the weaker claim that central syndicates delay liquidation, not that they produce exits. Gu et al. (2022) found that top-tier VCs cluster centrally and syndicate with each other persistently, which could reflect better coordination and support for their portfolio companies. This could also just reflect a preference for partnering with equally reputable investors and getting first look at the best deals. If it is the latter, centrality is a byproduct of who VCs choose to work with rather than a cause of anything. Putting these findings together, it creates an open question: if top-tier VCs cluster centrally and centrality predicts survival, it's unclear whether that reflects centrality doing actual predictive work, or centrality simply tracking reputation. Removing hub investors and watching what happens to prediction quality would be a way to tell the two apart, and neither does this. Bellavitis et al. (2017), for their part, found that cohesion effects depend on investor maturity and status, which goes against treating any single centrality measure as a universal predictor.

Startup-level outcomes

Traditional success definitions introduce measurement issues as exits are time-to-event outcomes that occur at specific points in time. Giot and Schwienbacher (2007) modeled

VC exits with survival analysis, treating IPOs, trade sales, and liquidations as separate hazards and handling right-censored firms properly. They concluded that exit hazards are inconsistent over the time of an investment and differ across exit routes. In the existing machine learning literature discussed in Section 3.3, censoring is often not considered. Success is instead collapsed into binary labels over a fixed period. Startups that simply haven't had time to exit yet get coded as failures, which understates true exit rates for younger cohorts. Secondly pooling IPOs and acquisitions throws away Giot and Schwienbacher's own finding that the two routes have different hazard profiles, and a lot of acquisitions with undisclosed terms are fire sales that return little capital. A binary exit label is therefore really predicting whether anything got recorded as an exit, not whether the outcome was an economic success.

3.2 Network View of VC Ecosystems and Startup Success

Representing the VC ecosystem as graphs

Beyond simply linking investors to one another and to startups, co-investment networks may convey information about trust, access, deal flow, and reputational spillovers. This pattern is observable in the tie data alone; yet since similarity-based selection, unobserved quality, and genuine spillovers can produce identical patterns, an identification problem arises (Shalizi & Thomas, 2011). The literature therefore shows at most that position is informative, not why it is informative. Esposito et al. (2022) asked whether success in VC-backed firms is explainable by relationship structure, constructing a bipartite network of startups and investors in the health sector. Their sample is limited to that sector, which has its own regulatory and capital-intensity quirks, possibly limiting its applicability to the wider market. They defined success three ways: success/failure, total funding raised, and funding trajectory. But funding raised and funding trajectory aren't independent of the network itself, since a startup's degree rises with each new investor, so a well-funded startup is by definition also a well-connected one. That makes two of the three success measures partly redundant with centrality. The third one is useful as it showed that connections to influential investors mattered more for success than local network structure did.

Startup networks change over time. Bonaventura et al. (2020) built a global startup-

to-startup network from employee mobility, treating firms as nodes and people transfers as edges. Network measures beat baseline models in out-of-sample tests, showing that simple network features carried predictive information which firm-level characteristics alone did not. However, by relying on employee-mobility edges and static centrality, they may capture short-lived hiring patterns rather than durable structural advantages.

3.3 Machine learning for startup success and exit prediction

Traditional machine learning for startup prediction

Traditional machine learning literature in the startup space treats the problem as a tabular classification. Prior studies find that no single classifier family dominates across problems, and that well-tuned mainstream classifiers often perform well against each other (Caruana and Niculescu-Mizil, 2006; Fernández-Delgado et al., 2014). Khamphukun and Narkbunnum (2026) reached the same conclusion for startups specifically, evaluating machine learning models on Crunchbase data and finding that evaluation and feature design, determine predictive performance. Hyytinen et al. (2023) built models to predict high-growth enterprises three years ahead, under a budget constraint mimicking a VC portfolio limit. Under their tightest screening budget, nearly 40% of the firms the model flagged went on to achieve high growth. That number is a positive predictive value, meaning most of the flagged firms fell short of high growth, and the measure says nothing about the high-growth firms the model missed entirely. High growth in a general enterprise population is very different from a venture exit, so the logic may not be directly transferable.

Predicting exit and success from non-network features

Exit prediction is a dominant topic within empirical papers; a central paper is Ross et al. (2021), which leverages machine learning for startup selection and exit prediction. Using a dataset of venture-backed firms, the authors developed CapitalVX, predicting whether a startup will exit via IPO or acquisition, remain private, or fail. The model had a predictive accuracy of around 88-90% out-of-sample for some outcome classes, not all. The exit classes are heavily imbalanced, so accuracy is not informative on its own, and ROC-AUC and PR-AUC would give a clearer picture.

Zbikowski and Antosiuk (2021) built ML models to classify firms as successful or unsuccessful using Crunchbase data, restricting themselves to features knowable at the moment of prediction so as to avoid look-ahead bias. They showed that naive ML models can pick up spurious correlations, and that careful data handling is needed to avoid misleading predictions. Their design avoids look-ahead bias by only using features knowable at prediction time. But country, region, and industry were the three most important features in their best model, so much of its predictive success just rides on regional and sectoral base rates rather than anything specific to the firms. This concern carries over to a graph-based design, since co-investment ties encode regional and sectoral clustering. Models in this space are trained on commercial databases such as Crunchbase, PitchBook, and Refinitiv, with coverage varying across providers. Kaplan and Lerner (2016) document the limitations and measurement challenges inherent in major venture capital datasets. Reporting to these databases is voluntary, and successful firms have more reason to keep their profiles updated than failed ones do. So a missing or thin firm record often means the firm failed. A model trained on this data can end up treating good data coverage itself as a sign of success. A missing or misclassified investor record may bias multiple features, resulting in a node or edge being removed or mislabeled in the co-investment graph. Across all three papers, data quality issues are treated as something to control for, leaving open whether they quietly shape the predictions themselves.

Limitations of non-network machine learning

Tabular-based ML models achieve high performance but have two key limitations that motivate a network-based approach. Firstly, investors are treated primarily as attributes to a startup, rather than as nodes in a larger system. This overlooks how the co-investment structure, including which other investors invest alongside and in what combinations, may influence information flow and access to follow-on capital. Secondly, these models rarely assess the stability of their predictions when the underlying structure changes. CapitalVX (Ross et al., 2021) and the model of Zbikowski and Antosiuk (2021) aim to maximize predictive accuracy but never examine how sensitive those predictions are to perturbations of that data. Graph representation addresses the first limitation by encoding relational structure, but does not address the second.

3.4 Graph-based Machine Learning

Graph-based prediction in financial networks

Network position appears to carry information that tabular features do not fully capture, raising the question of what these models are missing. Two distinct architectures emerge from the literature to address this.

Veličković et al. (2018) proposed graph attention networks (GATs). Rather than averaging every neighbor's features equally, as spectral convolution does, GAT lets each node learn how much weight to give each neighbor. A shared attention function scores every connection, so a neighbor with little useful information gets less weight. Applied to VC co-investment networks, which are sparse and hub-dominated, this weighting could allow the model to learn to discount a hub investor's failed deals and pay more attention to its successful ones. This is a partial fix as the weights are learned to reduce training error, not to reflect true outcome similarity. So if a hub's failed and successful investments look similar on the available features, the model still cannot tell them apart.

Hamilton et al. (2017) proposed GraphSAGE which learns an embedding function that samples and aggregates features from a node's neighborhood, enabling it to generalize to unseen nodes and even unseen graphs. In VC ecosystems, where new investors constantly enter, this inductive property makes the architecture more realistic. However, it still depends on the unseen node having a well-connected local neighborhood at inference time; newly entered or sparsely connected investors may not satisfy this condition.

Heterogeneous graphs

Zhou et al. (2020) surveyed the different families of graph neural networks and identified open issues in the field. They separate graph settings into two axes: homogeneous vs heterogeneous and static vs dynamic. Venture capital networks sit between both axes, as they are heterogeneous and they change over time. Zhang et al. (2021) apply heterogeneous graph learning to startup data, with their network containing startups, investors, and people. The SHGMNN framework learns startup representations and models label dependencies to predict high-potential startups,

improving predictive performance especially for early-stage startups with limited financial histories.

Graph learning applied to venture capital prediction

A small but growing body of literature is applying GNNs to VC/startup prediction. Zhang, Zhong, Ge and Xiong (2024) extend the SHGMNN framework detailed in Section 3.4.2 with VenGNN, adding person, VC-firm, institute, location and category nodes and using eleven meta-paths, including one that links VC firms through shared portfolio startups, to capture co-investment-style patterns indirectly. Lyu et al. (2025) built a bipartite network linking startups to investors and managers, incrementally updating graph embeddings (nodes) to predict startup success within 5 years of funding. Incrementally updated embeddings still assume the network is fully and accurately observed at each update step. Both papers focused on a fixed snapshot, with neither addressing what happens when the network is incomplete or noisy. Both results are consistent with earlier findings that adding relational structure improves performance, but only on the observed snapshot. Neither checks whether these gains hold under the structural changes discussed in Section 3.5.

3.5 Graph perturbation, adversarial robustness, and network stability

Adversarial attacks on graph structure

Trappolini et al. (2024) showed that graph neural networks remain vulnerable even when attacks are sparse. They introduced SAVAGE, an adversarial link prediction framework that maximizes attack effectiveness while minimizing the number of malicious nodes required, highlighting that GNN predictions were fragile to structural changes. This is different from naturally occurring changes like missingness and noise which are likely to be present in an imperfectly observed VC dataset. Zügner et al. (2018) introduced Nettack, an algorithm that modifies nodes' features and the graph's edges in a manner that is not statistically noticeable, yet caused classification accuracy to collapse, and these effects transfer across GNN architectures. Dai et al. (2018) trained a reinforcement-learning agent to choose which edges to add or remove so that a target node's classification flips, showing that structural attacks work even when the attacker cannot see the victim model's

gradients. In a VC-specific case this would be as if a firm simply disappeared from a deal record.

Zügner and Günnemann continued the research, examining whether a model can be proven robust (Zügner and Günnemann, 2019b). They developed a methodology to certify that a GCN's prediction for a node remains unchanged under any changes within a defined perturbation budget, and proposed a training procedure that quadruples the number of certifiably robust nodes without reducing accuracy. Certifying this is possible when using small, bounded perturbation budgets and GCN-style architectures, showing that robustness against small, bounded changes can be built into training itself.

Global and training-time attacks

Systematically missing, noisy, or structurally unreliable data can affect the training graph, preventing the inference graph from producing accurate results. Tao et al. (2024) raise this concern and improved robustness by refining the graph structure before message passing, training the model to recognize adversarial edges directly. This assumes labeled examples of corruption are available during training. That is a separate problem from identifying which real VC syndication ties are missing or misrecorded, so it is not directly applicable here. Sparse, real-world graphs like VC co-investment networks are especially vulnerable, as a small number of edge changes may represent a disproportionate shift.

Perturbation budgets and certified stability

Bojchevski and Günnemann (2019) extended this work to structural perturbations, adding or removing edges rather than changing node features. Structural changes are harder to certify because altering the adjacency matrix changes how information is normalized and aggregated across neighboring nodes. Their solution was a convex relaxation that gives a guaranteed upper and lower bound on a node's output score. They used this bound as a training objective to increase the number of certifiably robust nodes without reducing accuracy. This is not directly applicable to this study, as the certified bounds were derived for GCN-style architectures, not the GraphSAGE-based used here. Chen et al. (2025) extended the certification approach with a polyhedral-based abstract interpretation method, producing tighter bounds and

faster certification runtimes.

3.6 Synthesis and literature gap

Together, the literature shows that network position is associated with VC and startup outcomes, though the evidence in many cases is correlational. Tabular models, however, predict exits reasonably well without using network features or graph structure at all. This raises a specific concern for network features: the centrality measures they add are co-linear with reputation proxies, such as fund age, cumulative deal count, and past exit rate, that a well-specified tabular baseline already captures. The graph model may therefore be re-deriving structure the tabular model already knows, rather than adding anything new.

The perturbation and robustness literature in Section 3.5 also falls short here. It tests GCN-style architectures against adversarial attacks or certified bounds. Some assumes labeled corruption examples during training instead (Tao et al., 2024). None of it addresses naturally missing ties in a sparse, hub-dominated network like VC co-investment. The VC-specific GNN studies have a different gap. Zhang et al. (2021), Zhang et al. (2024), and Lyu et al. (2025) never test whether their predictive gains survive any structural change. No study reviewed here builds a VC co-investment network and benchmarks it against a tabular baseline with reputation proxies already included. None then tests whether any gain holds up once that network is perturbed. This study fills that gap. It tests whether co-investment structure adds predictive power to US startup exits beyond a well-specified tabular baseline, and whether any such gain remains stable under network perturbations.

4. METHODOLOGY

4.1 Empirical Design

The study deploys three models, which are scored using the same startups on the same train, validation, and test splits. A baseline tabular model with deal-level and firm characteristics only, a network-augmented tabular model with an additional centrality feature, and a heterogeneous graph neural network that reads the heterogeneous investor-startup graph. The baseline model is compared against the network-augmented model to

test H1, extending Sorensen (2007), Hochberg et al. (2007), and Bubna et al. (2020) by testing whether network position adds anything once firm and deal features are already in the model. The GNN is compared with the augmented tabular model to assess whether it can learn its own graphical representations, testing H2. The models are all then run through perturbation tests to test H3.

4.2 Baseline Models

For the baseline model set, 5 separate models are used: logistic regression, random forest, histogram-based gradient boosting, XGBoost, and LightGBM. Five models are used to reduce potential model bias, as each has its own assumptions about how features interact. Large-scale benchmarking studies find that no single classifier family dominates across problems and that well-tuned mainstream classifiers often perform comparably (Caruana and Niculescu-Mizil, 2006; Fernández-Delgado et al., 2014). Recent startup-outcome-prediction literature finds that evaluation and feature design, not algorithm choice, are the binding constraint on predictive performance (Khamphukun and Narkbunnum, 2026).

Each model is trained once using the baseline features (first round raised, cohort year, firm age at first round, sector), and a second time with the additional network features. Within the data sample, a class imbalance of 24.3% positive cases (successful exits) is present, which is handled during training by creating a class weight variable for the LogReg mode, HistGBM, and random forest (an option that scales up how much each minority-class example counts in the loss), and scale pos weight variable for the XGBoost and LightGBM (same idea as class weight). ROC-AUC and PR-AUC are used to evaluate the ranking quality threshold-free. ROC-AUC is the probability that a randomly chosen exiting startup is ranked above a randomly chosen non-exiting one with 0.5 being chance and 1.0 being perfect choice. PR-AUC is similar, but applies to precision-recall, and is more informative when there is a class imbalance.

4.3 Network-augmented Model

For the network-augmented model, two features are added to the baseline: mean_pagerank (the mean PageRank centrality of the startup's first-round investors) and missing_evc, a flag for startups whose first-round investors sit outside the graph's

largest connected portion. This aligns with Hochberg et al. (2007), which treats network position as one reputational signal rather than several. PageRank uses an 85% damping factor: a node's score mostly reflects its actual co-investment links, with the remaining 15% coming from random restarts so scores do not get stuck in disconnected parts of the graph. Esposito et al. (2022) adapt this from the damping-factor algorithm Brin and Page (1998) built for ranking web pages, applying it here to rank investors instead.

$$PR(i) = \frac{1-d}{N} + d \sum_{j \in N(i)} \frac{PR(j)}{\deg(j)} \tag{1}$$

PageRank formula: N is the investor node count, $N(i)$ is investor i's neighbors, j is one such neighbor, and $\deg(j)$ and $PR(j)$ are j's degree and PageRank score. The $(1-d)/N$ term gives every node its baseline random-restart score. The $PR(j)/\deg(j)$ term is the score neighbor j passes to i, split evenly across all of j's connections.

4.4 Heterogeneous GNN

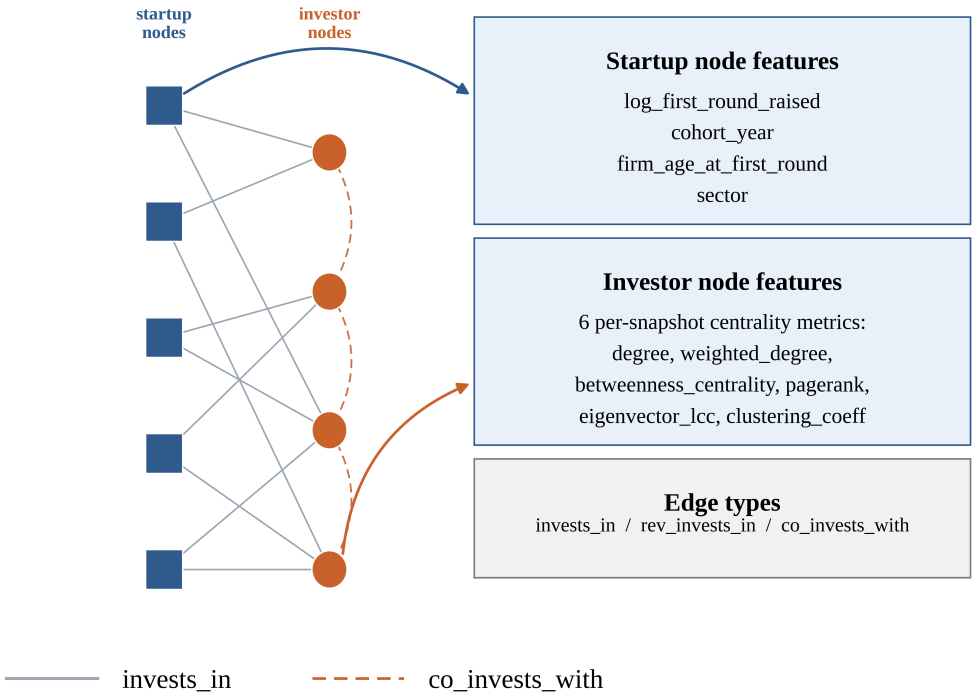


Figure 4.1: Heterogeneous investor-startup network, with the feature set stored on each node type

$$h_v^{(k)} = \sigma\left(W \cdot \left[h_v^{(k-1)} \,\|\, \mathrm{MEAN}_{u \in N(v)}\, h_u^{(k-1)}\right]\right) \tag{2}$$

HeteroConv update applied per relation: node v's previous embedding $h_v^{(k-1)}$ *and its neighbor mean* $\mathrm{MEAN}_{u \in N(v)} h_u^{(k-1)}$ *are concatenated, then passed through weights* W *and a ReLU nonlinearity* σ *to produce the next-layer embedding* $h_v^{(k)}$*.*

$$\hat{y}_s = \sigma\Big(W_{\text{head}} \cdot \mathrm{HeteroConv}_2\big(\mathrm{HeteroConv}_1(x, \mathcal{E})\big)_s + b_{\text{head}}\Big) \tag{3}$$

GNN forward pass: Where x *is the initial node features,* $\mathcal{E}$ *the edge types (invests-in, reverse-invests-in, co-invests-with),* $\mathrm{HeteroConv}_1$ *and* $\mathrm{HeteroConv}_2$ *the two SAGEConv layers, and subscript* s *selects the startup node's embedding, which* W_{head}*,* b_{head}*, and* σ *map to a predicted exit.*

A heterogeneous graph is a graph whose node set is partitioned into multiple types, with edges that can connect nodes of the same type or of different types (Zhou et al., 2020). In this research, the node types are investors and startups; edges connect investors to startups, representing investment relationships, and investors to other investors, representing co-investment ties. The GNN uses a two-layer heterogeneous architecture: a HeteroConv layer wraps a SAGEConv per relation. SAGEConv (Hamilton et al., 2017) is chosen because its inductive design handles investors who join after the training graph is built, the property that favors it over attention-based architectures such as GAT (Veličković et al., 2018), discussed in Section 3.4. It then aggregates across relations feeding the same node type before a linear head classifies each startup node (Figure 4.2). The graph has two node types (startup and investor) and three edge types (invests-in, its reverse, and co-invests-with), following the heterogeneous-graph approach of Zhou et al. (2020) and Zhang et al. (2021). Investor nodes receive 6 centrality metrics per snapshot, using only information known to investors at that time. The GNN uses PyTorch 2.12.0 and PyTorch Geometric 2.8.0. It was trained with the Adam optimizer for up to 200 epochs, with a weight decay term and early stopping after 20 epochs without validation ROC-AUC improvement, using one fixed random seed per hyperparameter configuration.

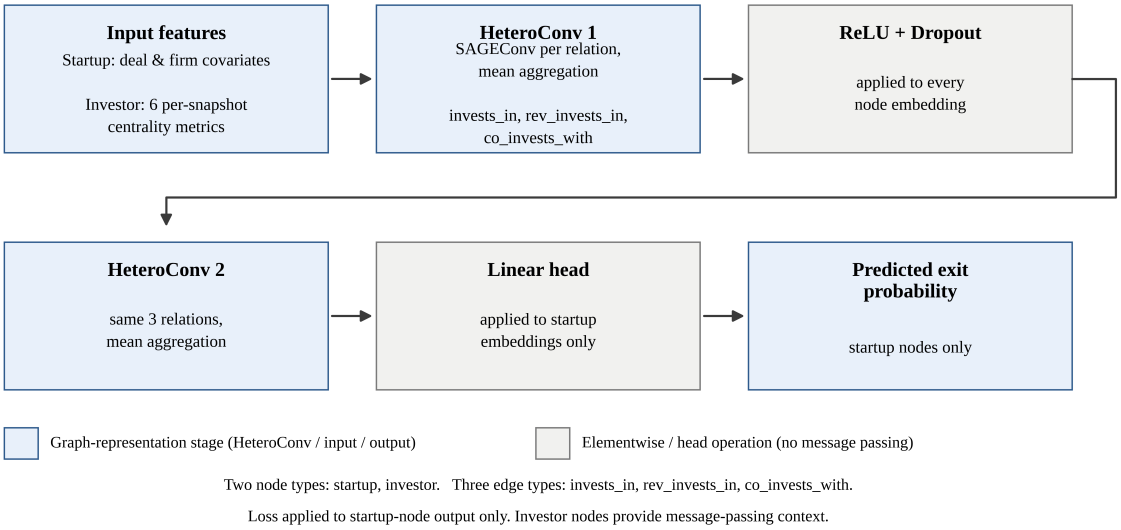


Figure 4.2: Two-layer heterogeneous GNN architecture

Train, validation, and test data are constructed in separate graphs rather than a single graph with masked nodes. GraphSAGE uses neighbor-sampling aggregation by design; hence a trained model does not need the graph it was trained on (Hamilton et al., 2017). A 12-combination grid search (training the model once for every combination of these settings, then comparing them) was performed over hidden dimensions (16, 32, 64), dropout (0.2, 0.4), and learning rate (0.001, 0.005); the best combination was selected based on validation ROC-AUC. Any class imbalance was corrected in the loss function using pos_weight (the GNN's own version of the same rebalancing approach, scaling up the loss contribution of exit cases during training).

The co-investment network is a weighted graph of investors, linking two investors when they invest in the same startup on the same date. Edge weight counts the rounds they have shared as of the graph's as-of date. Five metrics are computed per snapshot and aggregated (mean, max) across a startup's first-round investors (Table 4.1).

Table 4.1: Investor-level network metrics

Metric	**Definition**	**Source**
degree / weighted degree	number of distinct co-investment partners / sum of edge weights	General network measure
PageRank	Network PageRank, damping factor $\alpha = 0.85$	Esposito et al. (2022); Brin and Page (1998)
eigenvector centrality	computed within the graph's largest connected component only	General network measure
betweenness centrality	approximated via a $k = 200$-node sample for tractability	General network measure
clustering coefficient	local clustering coefficient (unweighted)	General network measure
missing_evc	indicator if investors do not lie in the snapshot graph's largest connected component	Esposito et al. (2022)

The co-investment layer builds one snapshot per split (2013-12-31 train, 2015-12-31 validation and 2017-12-31 test), with each startup scored against the same graph. A startup which is anchored in an early split can be scored against co-investment ties formed later within the same window. All the models run on a chronological split, anchored to each startup's first-round investment date, with training from 2010–2013, validation from 2014–2015, and test from 2016–2017. All 11 models scored the same 3,280 test startups.

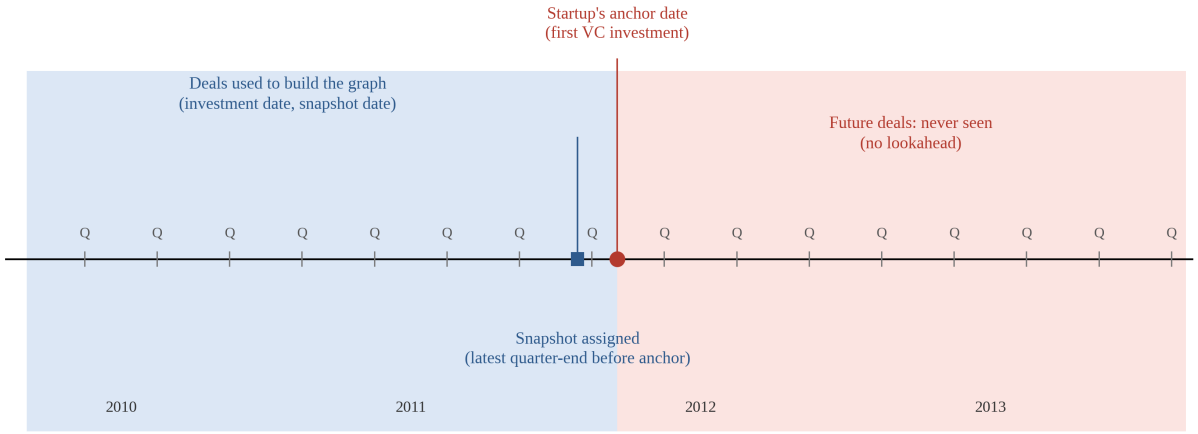


Figure 4.3: Quarterly snapshot assignment and the no-lookahead rule

The network is static, but is rebuilt every quarter to help reduce leaking future structure into early cohorts' predictions. Each startup is assigned the latest snapshot at or before its own anchor date, using only deals up to that point, following the look-ahead-bias principle set out by Żbikowski and Antosiuk (2021). At each startup's anchor date, only that startup's own deal-level attributes, its identified first-round investors, and the co-investment graph snapshot observed up to that date are available to the model; later-round investors, co-investment ties formed after the anchor date, and the exit outcome itself are excluded.

4.5 Perturbation framework

A single perturbation framework is applied to all models. Each mechanism is bounded to a fixed edge- or node-change budget, keeping the comparison across perturbation types fair (Zügner and Günnemann, 2019b). Both graph formats are converted into a common representation so the same perturbations apply consistently across models. Eight mechanisms are tested in total. Table 4.2 presents the five core tests: random edge deletion, preferential edge addition, degree-aware edge deletion, hub removal, and placebo removal. The three supplementary checks, random edge addition, edge rewiring, and node-feature noise (GNN only) were reported in Appendix C. These perturbations emulate real-world data-quality issues in VC databases, such as missing or spurious connections and missing investor records (Kaplan and Lerner, 2016). They test whether database incompleteness or quality issues prevent the models from reliably and consistently ranking

which startups are most likely to exit.

Table 4.2: Perturbation types and mechanisms (Applied to both models)

Perturbation type	**Mechanism**	**Budget / K (seeds)**
Random edge deletion	Edges removed uniformly at random	0.5–10% of edges (5 seeds)
Preferential edge addition	New edges added with probability proportional to the product of the two endpoints' existing degree	0.5–10% of edges (5 seeds)
Degree-aware edge deletion	Removal probability proportional to sum of endpoint degrees	0.5–10% of edges (5 seeds)
Hub removal	Deterministic removal of top K highest-degree investors	K = 1–25 (deterministic)
Placebo removal	Removal of K random, non-hub investors	K = 1–25 (5 seeds)



Figure 4.4: Perturbation and re-scoring pipeline

Random edge deletion represents uniformly missing co-investment ties. Preferential edge addition and degree-aware edge deletion are degree-sensitive, concentrating changes on already well-connected investors rather than spreading them evenly across the network. The supplementary Appendix C tests provide checks for uniformly spurious ties, degree-preserving rewiring, and noise in GNN node features. Hub removal runs separately; it removes a fixed set of the highest-degree investors and their edges, following the targeted-attack approach in Trappolini et al. (2024) and Zügner et al. (2018), both of which found sparse, targeted structural changes disproportionately degrade GNN predictions relative to random noise. A placebo test also removes the same number of random, non-hub investors, to confirm that any damage comes from centrality rather than node loss alone. Once perturbed, every graph has its investor features recomputed and is then passed

through the already-fitted model for re-scoring. The same procedure is used for both the tabular and GNN models. No model is retrained on any perturbed graph and every perturbation is applied only to the test-split graph after training is finished. This ensures that the model never gets the chance to learn to compensate for the damage. This isolates how an already-fitted model responds to structural change it never saw during training.

4.6 Stability metrics

Two kinds of stability are measured: aggregate (how much overall ROC-AUC shifts) and individual-level (how much each startup's own rank or classification shifts, via Spearman ρ and the flip rate; Table 4.3). The same method is used to measure the stability of both model types (tabular and GNN) on both clean and perturbed data. DeLong's test is used to see whether the difference in ROC-AUC between the two models is statistically significant. Bootstrap confidence intervals, built by resampling the test set, assess the same ROC-AUC difference (DeLong & Clarke-Pearson, 1988). Instead of a single p-value, they show the full range of values the difference could plausibly take. Testing eight comparisons on the same data increases the likelihood that one of them looks significant by luck. Two corrections are applied to the significance threshold to account for this: Bonferroni and Benjamini-Hochberg. Bonferroni raises the bar so high that even one false positive across all eight tests becomes unlikely. Benjamini-Hochberg (1995) is less strict, allowing for a small fixed rate of false positives among whatever results come out significant. This makes it better at catching real effects when they exist.

$$z = \frac{AUC_A - AUC_B}{\sqrt{\widehat{\mathrm{Var}}(AUC_A) + \widehat{\mathrm{Var}}(AUC_B) - 2\,\widehat{\mathrm{Cov}}(AUC_A, AUC_B)}} \tag{4}$$

$$p = 2\,(1 - \Phi(|z|)) \tag{5}$$

DeLong test formulas: AUC_A and AUC_B are the two models' ROC-AUC values on the same test set; their variances and covariance capture how noisy and correlated those scores are, and z converts the gap between them into a p-value via Φ.

Table 4.3: Stability metrics

Metric	**Definition**
ΔROC-AUC / ΔPR-AUC	Change in discrimination relative to the clean graph
Flip rate	Share of startups whose binary prediction, at a fixed 0.5 threshold, changes after perturbation
Spearman ρ	Rank correlation between clean- and perturbed-graph predicted probabilities (1.0 = identical ranking, 0 = no relationship)

5. DATA AND SAMPLE CONSTRUCTION

5.1 Data Sources

The analysis draws on Refinitiv/LSEG datasets covering US venture capital activity from 2000 to 2024, restricted to US-based companies. The datasets contain daily data for 51,464 US companies and 15,816 investor organizations (matched by Refinitiv's Firm Investor Organization ID). The "Undisclosed Firm" entry for investors is dropped throughout to avoid an artificial mega-hub. The datasets contain 21,961 logs of companies exiting via six exit types. Three are considered successful exits here: IPO, Merger, and Secondary Sales.

5.2 Sample Construction

To construct the investment sample, this study considers only the first investment for each company. Refinitiv's deal data starts in 2000, and the first investment is taken from Refinitiv's "Company First Investment Received Date" field, which covers a company's full history; the earliest deal record is used only as a fallback when that field is missing. Companies whose true first round predates their own deal history are excluded, and the sample is restricted to first rounds between 2010 and 2017, the only cohorts with a fully observable 7-year outcome window under the 2024–12–31 data cutoff (Table 5.1).

Table 5.1: Sample construction funnel

Stage	**N**	**% of total**
Total companies in deal data (2000–2024)	51,464	100.0%
True first round predates own deal history	-5,384	-10.5%
First round 2000–2009 (before entry window)	-13,127	-25.5%
First round 2018–2024 (7-yr window not over)	-21,156	-41.1%
Entry-window panel (2010–2017 cohorts)	11,797	22.9%
Inconsistent exit dates / right-censored rows	-6	-<0.1%
Classification sample	11,791	22.9%

5.3 Outcome Definition

The outcome is the timing and type of exit (IPO, acquisition/trade sale, liquidation, or continued private operation), given as a fixed-horizon binary label for comparability with prior graph-based VC prediction literature: exit within T=7 years of the first investment date. Despite the measurement concerns raised against binary exit labels in Section 3.1, this study still uses one because the GNN and tabular models compared need a single fixed-horizon outcome to classify. Exit dates come from a Refinitiv exit variable where available, and are otherwise coded as 'status-only', which are companies known to have exited but whose exit date is unknown. Status-only exits are conservatively coded as successes. Rows whose 7-year window is not yet observable and that show no exit are right-censored and dropped, rather than coded as failures by default. In the final sample (n = 11,791), 2,864 startups (24.3%) exit successfully within seven years, and 8,927 (75.7%) do not. There are 8,237 unique investors in the sample, with some investors repeating across data splits. The 7-year horizon and the right-censoring/dropping rule for not-yet-observable rows follow the survival-analysis treatment of VC exit timing in Giot and Schwienbacher (2007), who model IPO, trade sale, and liquidation as separate, time-dependent hazards rather than collapsing outcomes into a single fixed-period label.

The chronological train/validation/test split used throughout is set on the anchor date so that no outcome window or graph snapshot leaks across splits:

Table 5.2: Sample composition and co-investment graph size, by chronological split

Split	Anchor-date range	Startups	Exits (n, %)	Investor nodes	`invests_in` edges	`co_invests_with` edges
Train	2010 – 2013	5,317	1,379 (25.9%)	5,901	8,228	171,180
Validation	2014 – 2015	3,194	751 (23.5%)	6,785	5,362	193,728
Test	2016 – 2017	3,280	734 (22.4%)	8,237	6,061	224,112

5.4 Variable Construction

The predictors are calculated using information available at or before each startup's anchor date. Forward/backward selection dropped round size, deal value, round number, syndicate size, and stage/state categories as redundant. Every baseline covariate has precedent within the literature. The first-round-raised variable was taken from the CapitalVX paper by Ross et al. (2021), and the funding-history buckets were taken from Mishra et al. (2021). Cohort year and firm age were not taken from a specific paper, but are firm-characteristic and macroeconomic-context features listed as necessary by Mishra et al. (2021). Cohort year is the calendar year of the startup's first VC investment, used to group startups that entered the market around the same time. Sector follows the industry-classification variables of Esposito et al. (2022) and Żbikowski and Antosiuk (2021), who explicitly mention the need for an industry-composition feature to avoid biasing the models. The anchor date is the date at which the startup received its first investment. The network features, mean PageRank and missing evc, are used for the tabular models as a full centrality feature, as in Esposito et al. (2022).

The variables in Table 5.3 were mostly unadjusted. Log First Round Raised was logged because raised amounts are heavily right-skewed, with a small number of large rounds otherwise dominating the distribution. Firm age at first round was winsorized at the 1st and 99th percentiles, capping its range at 0–22.55 years. This removes the influence of implausible values arising from data quality issues in the underlying company_founded_date variable, consistent with the coverage and measurement limitations Kaplan and Lerner (2016) document for commercial VC databases.

Table 5.3: Descriptive statistics ($n = 11{,}791$)

Variable	**Type**	**N valid**	**Mean**	**Median**	**Std**	**Min**	**Max**
Exit Within T	Binary (dependent)	11,791	0.243	0	–	0	1
Mean PageRank	Continuous	10,240	0.00079	0.00035	0.0011	0.00002	0.00810
Missing EVC	Binary (coverage flag)	11,791	0.163	0	–	0	1
Log First Round Raised	Continuous (log)	11,791	1.148	0.993	1.011	0.000	7.004
Firm Age at First Round	Continuous (years)	11,791	3.44	2.15	5.00	0	22.55
Cohort Year	Discrete (2010–2017)	11,791	2013.8	2014	2.25	2010	2017
Sector	Categorical (13 levels)	11,791	–	–	–	–	–

Table 5.4: Variable definitions

Variable	**Definition / formula**	**Treatment of missing**	**Source**
Log First Round Raised	$\ln(1 + \text{first_round_raised_mn})$, first_round_raised_mn = Round Equity Total of the anchor-date deal record(s)	filled with 0	Ross et al. 2021 & Mishra et al. 2021
Cohort Year	calendar year of anchor_date	N/A	Used as "macro-economic context indicator" as described by Mishra et al. 2021
Firm Age at First Round	(anchor_date - company_founded_date) / 365.25	filled with 0	Falls under the "firm characteristics" bucket in Mishra et al. (2021)
Sector	Refinitiv TRBC Economic Sector, one-hot encoded (13 categories observed)	coded "Unknown"	Ross et al. 2021 and Żbikowski & Antosiuk 2021
Mean PageRank	mean PageRank ($\alpha = 0.85$) of the startup's first-round investors, on the pre-anchor snapshot graph	filled with 0	Esposito et al. 2022 and Hochberg et al. 2007
Missing EVC	1 if none of the startup's first-round investors lie in the snapshot graph's largest connected component, else 0	N/A	Esposito et al. (2022)

5.5 Feature Selection Ablation

To arrive at the six-variable set for the tabular models, three rounds of eliminations were run. Initially, five raw investor centrality measures (degree, weighted degree, eigenvector, betweenness, clustering) were reduced into a single summary feature (mean PageRank), because the five measures were highly collinear, aligning with Esposito et al. (2022). Secondly, three literature-motivated centrality features were tested: own_degree (a startup's own bipartite degree, following Esposito et al. 2022), tie_strength (mean weighted degree of an investor's repeat co-investment ties, following Bellavitis et al. 2020), and community (Louvain community share, following Bubna et al. 2020 and Gu et al. 2022). None of the features meaningfully improved ROC-AUC over the existing feature set; therefore, they were not included.

Of the investor-level metrics computed per snapshot, only mean PageRank and Missing_evc survived into the final feature set. Missing_evc is included in the model because PageRank fails to recognize peripheral but connected investors; it gives these investors visibility as well.

Table 5.5: Model features. Startup and deal-level features are identical across all three models; investor and co-investment network features differ by model.

Feature	**Baseline tabular**	**Network-augmented tabular**	**GNN (heterogeneous)**
log_first_round_raised	Yes	Yes	Yes
cohort_year	Yes	Yes	Yes
firm_age_at_first_round	Yes	Yes	Yes
sector	Yes	Yes	Yes
mean_pagerank (aggregated, startup-level)	–	Yes	–
missing_evc (aggregated, startup-level)	–	Yes	–
degree / weighted degree (raw, investor node)	–	–	Yes
PageRank (raw, investor node)	–	–	Yes
eigenvector centrality (raw, investor node)	–	–	Yes
betweenness centrality (raw, investor node)	–	–	Yes
clustering coefficient (raw, investor node)	–	–	Yes

6. RESULTS

6.1 Baseline, Network-Augmented, and GNN Model Performance

Table 6.1: All eleven models, clean-graph test performance (GNN Δ and DeLong p are reported against the best baseline model for GNN (RandomForest) and against their own baselines for the tabular models (3,280-startup test set)

Model class	**Model**	**ROC AUC**	**95% CI**	**PR AUC**	**Brier**	**Δ vs. baseline**	**DeLong p**
Baseline	RandomForest	0.6489	[0.626, 0.672]	0.3503	0.2440	–	–
	HistGBM	0.6457	[0.623, 0.668]	0.3318	0.2475	–	–
	LogReg	0.6450	[0.622, 0.668]	0.3445	0.2053	–	–
	XGBoost	0.6434	[0.621, 0.666]	0.3333	0.2482	–	–
	LightGBM	0.6408	[0.618, 0.663]	0.3311	0.2446	–	–
Network	LogReg	0.6525	[0.630, 0.675]	0.3454	0.2001	+0.0074	0.035
augmented	RandomForest	0.6464	[0.624, 0.668]	0.3335	0.2308	-0.0025	0.728
	LightGBM	0.6434	[0.621, 0.665]	0.3217	0.2279	+0.0026	0.721
	HistGBM	0.6431	[0.621, 0.665]	0.3244	0.2351	-0.0026	0.714
	XGBoost	0.6428	[0.621, 0.665]	0.3238	0.2351	-0.0007	0.917
GNN	Heterogeneous GNN	0.6582	[0.636, 0.680]	0.3468	0.2072	+0.0093	0.270

Among the baseline models, the RandomForest model had the highest ROC-AUC (0.6489, 95% Confidence interval [0.626, 0.672]) and PR-AUC (0.3503) of the 5 baseline models (Table 6.1). All the baseline models were above the 0.5 threshold, and are tightly clustered, with a range of just 0.008 in ROC-AUC separating the best from the worst models. Adding mean_pagerank and missing_evc shifts the ROC-AUC for the models between -0.0026 and +0.0074. Some models (LogReg and LightGBM) show a small uplift, while others (RandomForest, HistGBM, XGBoost) show a sign flip (worse performance) (Table 6.1).

The lift in LogReg is also nominally significant, with a DeLong p-value of 0.035; the remaining models have values between 0.6 and 0.92, indicating they are indistinguishable from zero. This test set includes eight comparisons. When correcting for multiple comparisons using the Bonferroni or Benjamini-Hochberg methods, no

models survive. This is interpreted as network features not improving discrimination between firms that exit and those that do not. For calibration (the mean squared difference between predicted and actual exit probability with 0 being perfect and 0.25 being an uninformed guess), the Brier score decreased from 0.2053 to 0.2001 for LogReg, confirming that the network-augmented model's predicted probabilities track actual outcomes more closely.

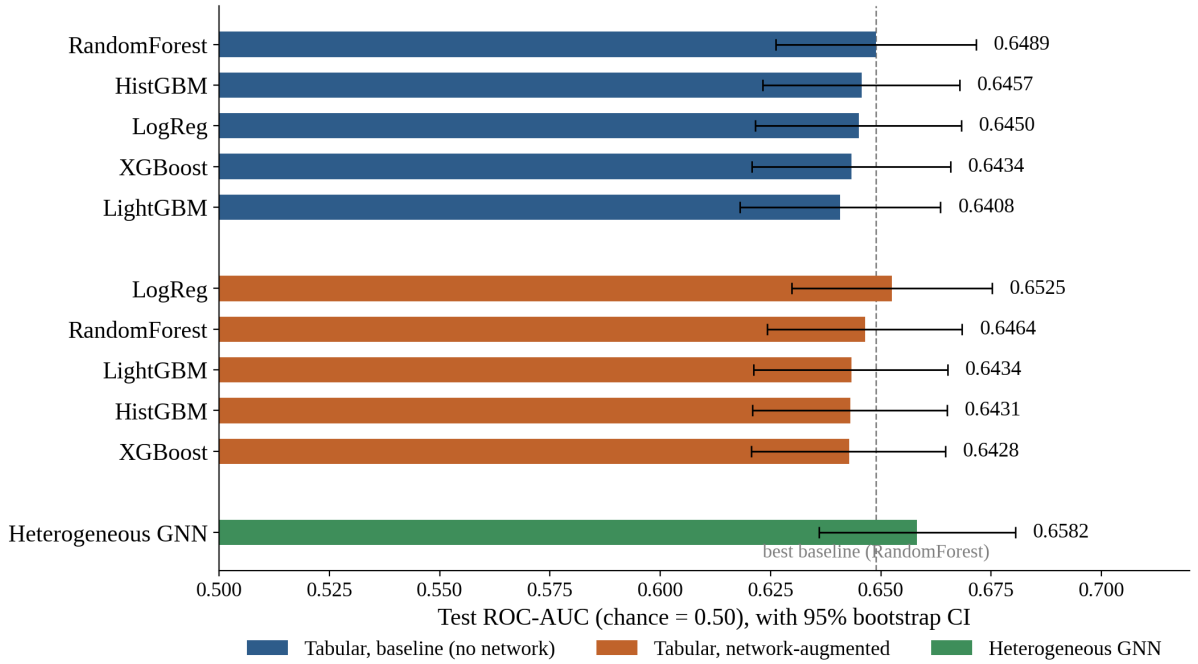


Figure 6.1: All eleven models, clean-graph ROC-AUC by specification (baseline, network-augmented, and GNN)

The GNN produced the highest point estimates among all eleven models, with ROC-AUC of 0.6582 and PR-AUC of 0.3468 (95% CI [0.636, 0.681]), selected from a 12-point validation grid (hidden_dim = 64, dropout = 0.4, lr = 0.001, validation ROC-AUC 0.6787). That advantage was not significant when compared against the best baseline (RandomForest). The GNN's ROC-AUC uplift of +0.0093 carried a DeLong's p of 0.27, and against the best network-augmented model (LogReg), an uplift of +0.0058 carried a p of 0.34. All eleven models fall within a 0.017 ROC-AUC range, and their 95% CIs overlap heavily (Table 6.1; Figure 6.1). Given a test sample of 3,280 startups, the study is underpowered to detect differences of this size, and the nominal uplift from the network-augmented models does not survive correction for multiple comparisons. Within the power this test could detect, network position does not add predictive value for the

tabular models and the GNN. These results do not support H1 and H2.

Table 6.2: Edge-level perturbations, summary across all 5 network-augmented models and GNN, 1% and 10% budgets. Mean is the average across all perturbation budgets.

Perturbation type	**Budget**	**Mean ΔROC-AUC**	**Worst ΔROC-AUC**	**Mean flip rate**	**Worst flip rate**	**Mean Spearman ρ**
Random edge deletion	1%	+0.00012	-0.00264 (LightGBM)	4.16%	7.51% (GNN)	0.9840
	10%	+0.00016	-0.00250 (LightGBM)	4.15%	7.07% (GNN)	0.9835
Preferential edge addition	1%	+0.00010	-0.00276 (LightGBM)	4.17%	7.53% (GNN)	0.9841
	10%	+0.00041	-0.00206 (LightGBM)	4.18%	7.37% (GNN)	0.9839
Degree-aware deletion	1%	+0.00020	-0.00246 (LightGBM)	4.16%	7.46% (GNN)	0.9841
	10%	+0.00006	-0.00266 (LightGBM)	4.28%	6.99% (LightGBM)	0.9827

Table 6.2 and Figure 6.2 report the three edge-level perturbation results, random deletion, preferential addition, and degree-aware deletion, at the 1% and 10% budgets. Adding random deletion, degree-aware deletion, and preferential edge addition had minimal impact on all the models tested. Throughout, the mean change in ROC-AUC across the six models stayed within a narrow band, never exceeding +0.00041 at any single budget. Spearman ρ never fell below 0.984 at the 1% budget, and only dips to 0.9827 at the 10% budget of degree-aware deletion (Table 6.2). The worst result in this grid was LightGBM under preferential edge addition at the 1% budget, where ROC-AUC fell by -0.00276. Preferential edge addition tested whether a more realistic, degree-weighted mechanism for spurious syndication ties was more damaging, but it wasn't, leaving ROC-AUC only marginally affected (+0.00041).

The supplementary checks in Appendix C tell a similar story, except for random edge addition at the 10% budget, which produced the largest ROC-AUC drop in the study (-0.0045, LightGBM). Node-feature noise had a near-zero impact on the GNN alone (-0.00008 at the 10% budget, flip rate under 1%), showing that its higher flip rate under structural perturbation is not a general property of the architecture, but a result of changes to the graph itself. Although these perturbations had minimal impact on

ROC-AUC, the GNN's flip rate was 1.5–2 times higher than any tabular model's when randomly perturbed (Figure 6.2).

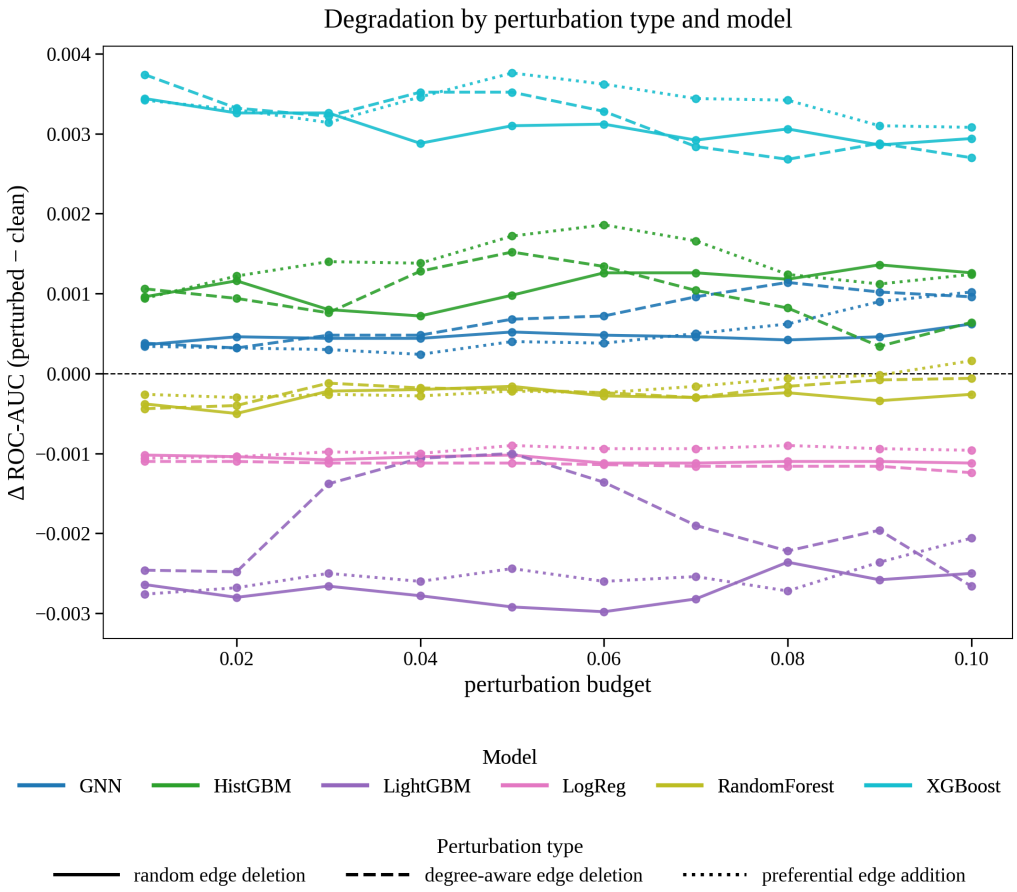


Figure 6.2: Degradation by perturbation type across budgets, GNN and the five network-augmented tabular classifiers

6.2 Hub-removal results

Hub removal strips the top-K most connected investors from the graph, one at a time, up to K=25. At K=25, LogReg is the most robust model, losing just 0.0077 ROC-AUC. Spearman ρ across all six models ranges from 0.92 to 0.97. The GNN's raw ROC-AUC looks strong, with a drop of only 0.0134 at K=25 and, at K=5 and K=10, it beats every tabular model. That number is deceptive, as at every K the GNN's flip rate is the worst of any model, reaching 11.16% at K=25, nearly double the average across the five tabular models. Its Spearman ρ is 0.923 at K=25, the lowest of the six, meaning its individual predictions move around more than any other model even as its aggregate ROC-AUC holds up.

Table 6.3: Hub removal (top-K most-connected investors removed), anchor budgets, (Uses Network augmented tabular models)

Model	**K=5 ΔROC-AUC**	**K=10 ΔROC-AUC**	**K=25 ΔROC-AUC**	**K=25 Spearman ρ**	**K=25 flip rate**
LightGBM	-0.0026	-0.0080	-0.0171	0.9268	8.78%
RandomForest	-0.0005	-0.0056	-0.0153	0.9344	6.10%
HistGBM	-0.0002	-0.0058	-0.0151	0.9376	6.01%
XGBoost	-0.0008	-0.0052	-0.0139	0.9439	5.40%
LogReg	-0.0010	-0.0029	-0.0077	0.9721	2.77%
GNN	+0.0011	-0.0027	-0.0134	0.9230	11.16%

Observing Figure 6.2, LightGBM is the single worst-affected model on raw ROC-AUC at every K from 6 through 25. Its degradation curve separates from the other tabular models by around K=6 and never rejoins them. Second, the GNN's flip rate is proportionally the largest compared to the tabular pack at the smallest budgets, and the gap narrows as K grows. At K=1, the GNN's flip rate (7.96%) is more than eleven times the mean of the five tabular models (0.70%), a gap that closes to under two times by K=25 (11.16% vs. 5.81%). The GNN is therefore more distinctly unstable compared to the tabular models when only a handful of hubs are gone. In a realistic setting, this means it's more prone to performance degradation when an investor record is missing.

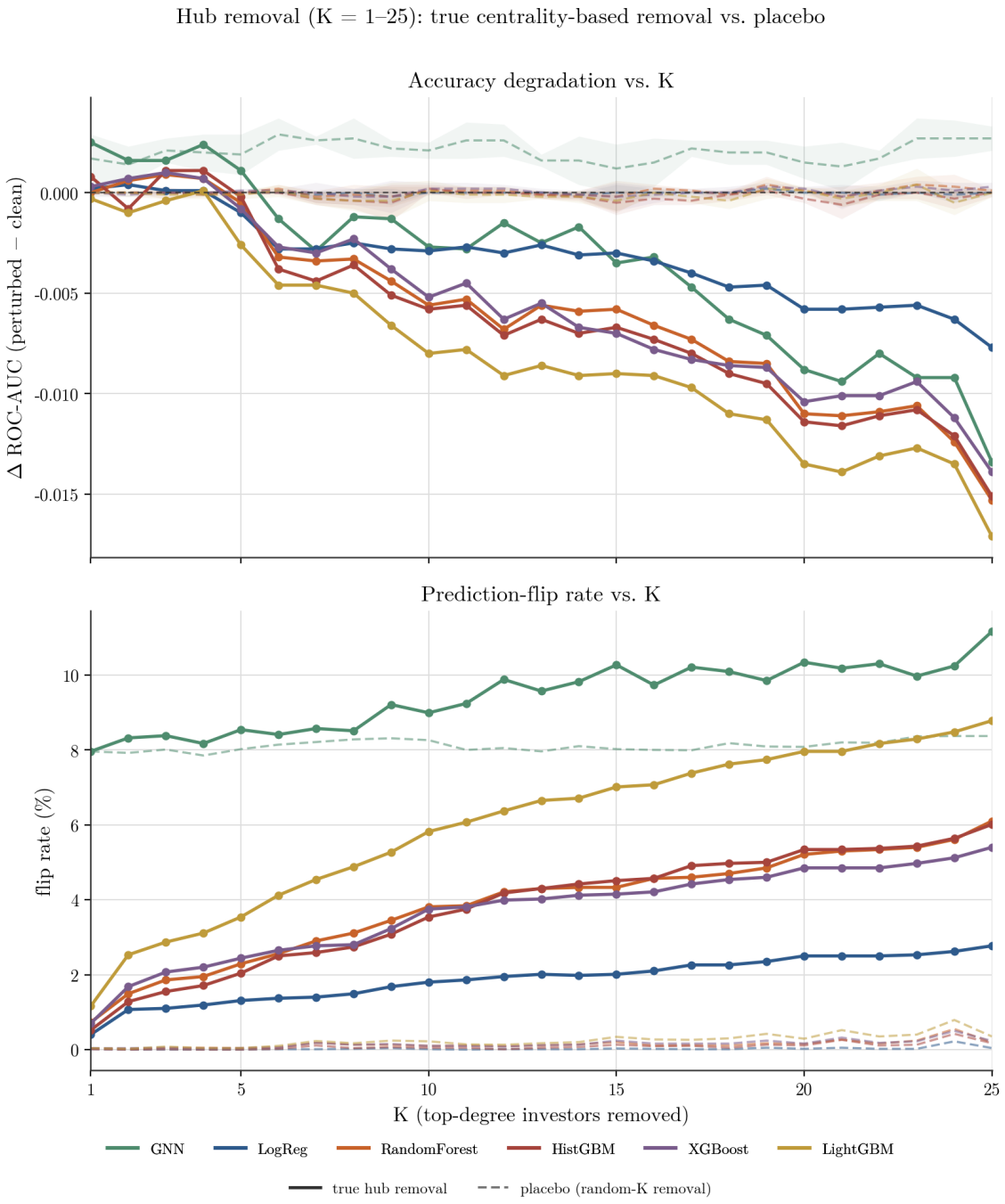


Figure 6.3: Hub removal at K = 1–25, true centrality-based removal (solid) vs. placebo random-node removal (dashed, ± 1 SD band)

A further placebo test checked whether removing K random non-hub investors would damage the model to the same degree as removing the top K investors. Using the same K=1–25 grid, true hub removal beat the placebo in 142 of 150 (model, K) cells (94.7%). All eight exceptions cluster at low K (mostly the GNN, plus one isolated cell each for LightGBM and RandomForest), are within noise ($\leq$ 0.0004 ROC-AUC), and disappear entirely from $K \geq 13$ onward, where true hub removal beats the placebo in

all 78 remaining cells. The placebo's change in ROC-AUC never exceeded +0.0003 for the five tabular models and +0.0027 for the GNN, far smaller than the drop from true hub removal. This confirms that the damage comes specifically from removing the most connected investors, centrality is what drives network fragility. The five tabular models' placebo flip rates stayed under 0.4%. The GNN is the outlier on flip rate as its placebo flip rate reaches 8.37% at K=25, close to the 11.16% seen under true hub removal. This suggests that the GNN's individual predictions are sensitive to losing any well-embedded investor, not only the most central ones.

Table 6.4: Placebo test, true hub removal vs. random-node removal (K=25)

Model	**True hub ΔROC-AUC**	**Placebo ΔROC-AUC (mean)**	**True hub flip rate**	**Placebo flip rate (mean)**
LightGBM	-0.0171	+0.0000	8.78%	0.35%
RandomForest	-0.0153	+0.0001	6.10%	0.19%
HistGBM	-0.0151	+0.0000	6.01%	0.17%
XGBoost	-0.0139	+0.0003	5.40%	0.23%
LogReg	-0.0077	+0.0000	2.77%	0.04%
GNN	-0.0134	+0.0027	11.16%	8.37%

Hypothesis 3 is supported for targeted, centrality-based perturbation, as removing the network's most-connected investors produces the largest effect of any perturbation tested on every model. There is a divergence between aggregate- and individual-level stability for the GNN. Its ROC-AUC barely moved at any K, but its flip rate is consistently the highest of any model, so aggregate accuracy masks this failure.

7. DISCUSSION

7.1 Interpretation of Findings

Hypothesis 1 did not hold as the boost from network features is not statistically significant. Only LogReg cleared the bar, with a DeLong p of 0.035. That is the outlier out of eight comparisons on the same test set, close to what chance alone predicts at $\alpha = 0.05$ (a 1-in-20 false-positive rate). Hochberg et al. (2007) and the broader literature in Sections 3.1 and 3.2 treat network position as informative for portfolio performance. These results do not contradict that literature so much as align with Esposito et al. (2022). In practice,

the mean-PageRank-style centrality feature behaved as noise rather than signal for the tabular models. It moved point estimates but produced nothing that survives correction for multiple comparisons.

Hypothesis 2 is also not supported by the results. Although the GNN produced the highest point estimate among all eleven models, that advantage did not reach statistical significance against either the best baseline or the best network-augmented model. Together with the flip-rate results discussed later, this suggests the GNN's slight ROC-AUC edge isn't extra predictive signal. It looks more like the effect of a different inductive bias applied to the same startup-level information: a small, test-specific ROC-AUC gain traded for less stable rankings. A richer dataset or better-engineered node and edge features may be needed before the GNN's structural sensitivity translates into a predictive advantage over the tabular baselines.

Hypothesis 3 offers an interesting insight as the models were robust to random and structural edge perturbations but fragile to targeted removal. This is similar to the findings of Trappolini et al. (2024), which found that sparse, targeted attacks disturb model outputs far more than random noise does. The GNN illustrates this best. As shown in Section 6.2, the GNN's aggregate ROC-AUC held steady under hub removal while its flip rate more than double the tabular mean. The GNN's Spearman ρ fell to 0.923, the lowest of any model tested. Practically, the GNN's estimate of how many startups will exit barely moves, but its opinion on which specific startups will exit shifts substantially. A model can therefore hold its aggregate score steady while reordering a meaningful share of the startups it ranks. This means ROC-AUC alone cannot certify that a graph-based model's outputs are stable enough to rely on operationally. Any evaluation of a GNN intended for real-world ranking should treat rank-stability metrics such as flip rate and Spearman ρ as a required complement to ROC-AUC.

The preferential-edge-addition result reported above (Figure 6.2) is a useful negative finding for interpreting Hypothesis 3 as the models are not vulnerable to "realistic-looking" noise, but rather only to noise that removes or restructures the specific high-degree nodes the network features are built on. Preferential edge addition directs most of the spurious (fake) ties toward investors who are already well connected, rather than spreading them evenly across the network. That barely changes those investors' neighborhoods, since a few extra fake ties are negligible against an already-large degree. The damage documented

below instead depends on where in the degree distribution the noise falls, low-degree investors are affected far more by the same number of spurious edges, regardless of how plausible the noise-generating mechanism is.

The hub removal K = 1–25 grid reported above suggests that the GNN's distinctive instability is not due to the perturbations being large, but rather that they were targeted. This issue is most pronounced where real-world data-quality issues are likely to occur, such as a single missing or misrecorded investor record. But this could be interpreted differently. The most connected investors might just be the ones with the most complete records, not the most important ones structurally, since private-market reporting is voluntary. The placebo test rules this out: removing the same number of random, ordinary investors barely moves the model, while removing the true hubs does real damage. So it can be inferred that centrality itself is driving the fragility, not which investors happen to have complete records. In practice, that means a handful of departures, acquisitions, or reporting gaps among a small set of well-connected investors can shift rankings more than losing the same number of random investors ever would.

Table 7.1: Hypotheses summary

Hypothesis	**Verdict**	**Key evidence**
H1: network position adds predictive value	Not confirmed	0 of 8 headline comparisons significant after correction; LogReg +0.0074, $p = 0.035$ uncorrected only
H2: GNN beats hand-engineered network features	Not confirmed	GNN 0.6582 highest point estimate; vs. best baseline $p = 0.27$, vs. best network model $p = 0.34$
H3: robustness depends on perturbation type	Confirmed	Random noise mean ΔROC-AUC within ±0.0005 at 10% (individual models range wider); hub removal ($K = 25$) down to -0.0171, Spearman ρ to 0.92; true hub beats placebo in 142/150 cells across the full $K = 1$–25 grid

Why Network Structure Adds Little Predictive Value

A possible reason the GNN barely reacted to edge-level perturbations is that the graph structure carried little predictive weight in the first place. This lines up with the concern raised in Section 3.4.1 that connected nodes are assumed to share similar outcomes. A hub investor's ties span both successful and failed ventures, so the signal carried by any single edge is mixed rather than a clean marker of outcome similarity (Veličković et al., 2018). Random edge perturbations, such as additions, deletions, and rewiring, left the aggregate metrics largely undisturbed. The model's predictions rested mostly on node-level information.

Similarly, in the tabular network features, investor centrality carries limited independent information once deal-level features were in the baseline specification. Centrality measures tend to be highly collinear with each other (Esposito et al., 2022). Investors and startups choose their own ties, so network position is not random. That self-selection makes it hard to tell whether centrality reflects a genuine network effect or just unobserved investor and startup quality. This mirrors the sorting-versus-influence problem raised for Sorensen (2007) in Section 3.1. Gu et al. (2022) raised a comparable concern about top-tier VCs clustering centrally, since that pattern could just reflect a preference for partnering with equally reputable investors. Well-connected investors may select into startups whose quality is already visible through other channels, or those startups may select them.

The network's predictive signal can be weak on average while still being concentrated in a small number of hub investors. Most of the network's predictive power could sit with a small number of well-connected hub investors, concentrated rather than spread evenly across every tie. So the typical tie could carry very little information while a few specific nodes carry a lot. Removing 25 random nodes leaves most hub neighborhoods largely intact. Removing the 25 hubs directly removes the nodes carrying that concentrated signal.

The placebo test demonstrates this directly. A model relying only on noise would degrade by roughly the same amount whether the top hubs or an equal number of random investors were removed, since noise spreads evenly across the network. True hub removal instead beat random-node removal in 142 of 150 (Table 6.4), showing a

concentrated network signal exists. This concentrated signal is what makes network position structurally important, distinct from its contribution to aggregate accuracy.

7.2 Practical Implications

Hub investors are structurally important even though neither their network features nor the GNN add measurable predictive lift on their own. A small set of highly connected investors is where the graph's risk concentrates. Random noise from missing data, missing syndicate members, or misclassified investor records has little effect, but losing a few core hub investors significantly damages the model. That is the core argument for consistent data monitoring, since a single missing node, investor or startup, can distort the model's outputs.

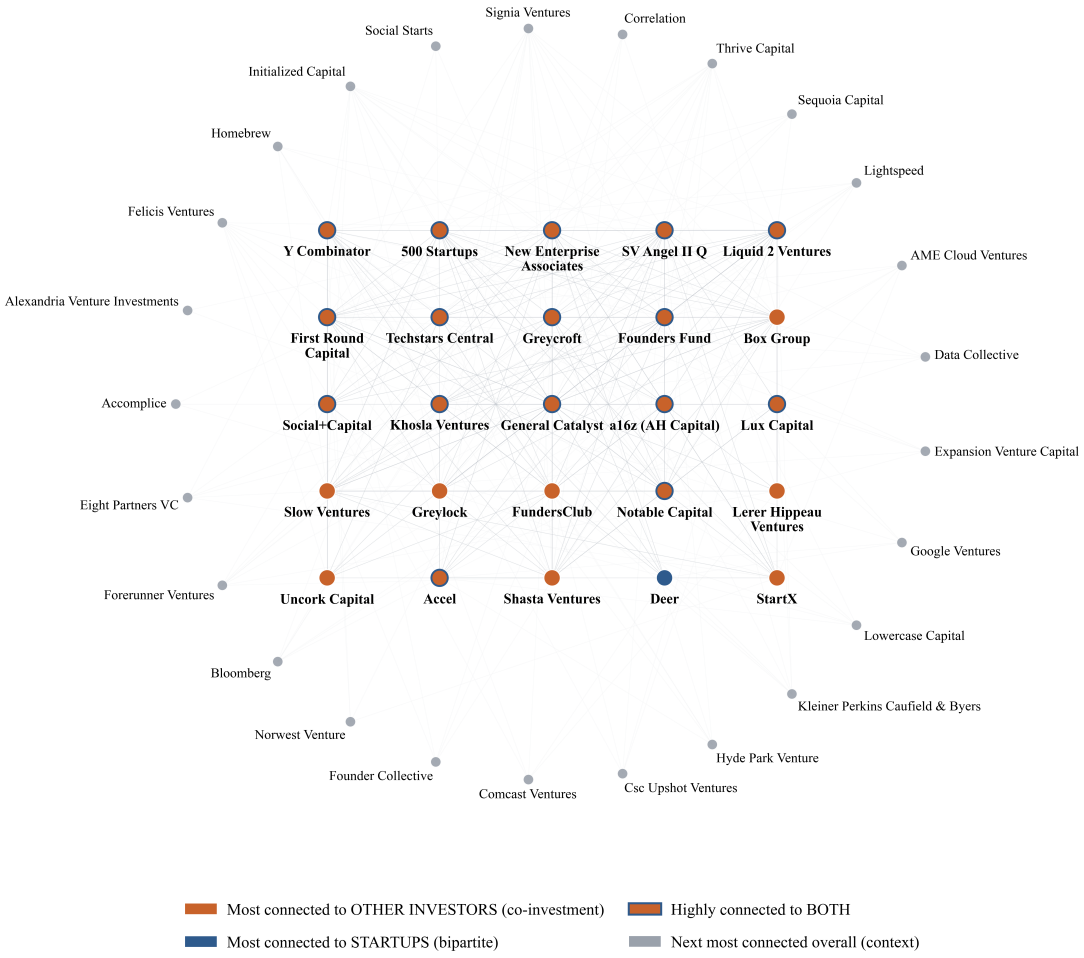


Figure 7.1: Top 25 most connected first round Investors

This split between stable accuracy and unstable ranking matters differently

depending on where in the investment process the model is used. In screening, a startup can drop off a shortlist, or a previously unranked one join it, simply because a hub investor left the graph. In due diligence, a rank that shifts depending on which graph snapshot is produced. This isn't telling you anything meaningful about the startup itself. Portfolio construction is exposed even more directly, since relative rank decides capital allocation. Just 0.16% of investor nodes sit behind 9.2% of all first rounds, so losing the wrong node carries outsized weight. The governance stakes follow from the same fact, as a model can pass every aggregate accuracy check and still reorder more than one in ten predictions after a routine data refresh. Flip rate and Spearman ρ catch that; aggregate accuracy alone would not.

7.3 Limitations

The design here cannot be extended to portfolio strategy, as mean PageRank cannot tell whether a well-connected investor drives better outcomes or simply selects startups already likely to succeed. Another design limitation involved the chronological split between the 2010–2013 train, the 2014–2015 validation, and the 2016–2017 test, which aimed to prevent lookahead. This may have made the data susceptible to market conditions present during those periods. The small test sample, only 3,280 startups, likely compounded these design issues. With 95% bootstrap confidence intervals on ROC-AUC roughly ± 0.02 to ± 0.025 wide around each point estimate, the test set had limited power to detect small differences between models, which is a more likely explanation for the insignificant DeLong p-values than a genuine absence of any difference.

The design choices made introduced several scope limitations. The dataset was restricted to US-based companies, hence the findings may not apply to other VC markets with different syndication norms. The analysis also faced coverage and measurement limitations common in commercial VC datasets, as Refinitiv investor records may not have been perfect, which partly motivated the robustness checks against missing or misclassified investors. The network and features were also fixed to each startup's first VC round. Later-round investors and co-investment ties were not observed. The fixed seven-year exit window simplifies the time-to-event process. The fixed horizon improves comparability with the graph-based VC prediction literature,

but it fails to capture how exit timing itself varies. The binary exit framing trades the richer hazard information from Section 3.1 for comparability with the GNN literature. Similarly, the perturbation tests apply structural change to a single fixed snapshot of the test graph, not to a network evolving in real time, so they show sensitivity to that one kind of change rather than to gradual, real-world attrition. Finally, this study tested only one heterogeneous GNN architecture (HeteroConv), so the robustness findings describe this architecture's behavior, not GNNs as an entire model class.

7.4 Future Improvement

Moving forward, refitting the models across successive training windows and then scoring each against an out-of-sample block would show whether the H1/H2 results and the size of removal effects were genuine or driven by the single 2016/2017 test set used. Applying Zügner and Günnemann's (2019b) and Bojchevski and Günnemann's (2019) certification methods would help create a provable bound for perturbation budgets. The network features used did not show a causal effect of network position. This would require examining investors' strategies and instruments. Future work could also extend the preferential-versus-random perturbation comparison introduced here to the hub-removal setting itself. Testing whether a degree-weighted rather than purely rank-based removal order changes the point at which damage accelerates. A test could also be conducted to determine whether the $K \geq 6$ divergence point identified for LightGBM is a general property of tree-based tabular models or specific to that implementation.

8. CONCLUSION

This study aimed to investigate whether venture capital exit predictions derived from a heterogeneous co-investment network were robust to perturbations and whether the GNN added predictive value beyond tabular models. Three hypotheses were designed to test these claims, evaluating a cohort of 11,791 startups between 2010 and 2017. The three hypotheses are congruent in the story they tell. Network position carries at most a small, statistically weak signal for exit prediction, and that signal does not survive multiple comparison corrections.

Hypotheses 1 and 2 were not confirmed Table 7.1. Even so, GNNs are worth pursuing

further in venture capital applications, Zhang et al. (2021) and Lyu et al. (2025) both find heterogeneous graph learning outperforming feature-based baselines in similar settings. The strongest contribution is Hypothesis 3, where every model is largely unaffected by random perturbations but damaged by targeted ones, such as hub removal. Section 6.2 shows that the damage is most visible in rank stability rather than aggregate accuracy.

Together, Hypotheses 1 through 3 point toward the same conclusion. Network position, as captured by these features, adds little predictive power under normal conditions. But the placebo comparison in Section 6.2 shows it does structural work. True hub removal degrades every model far more than removing an equal number of random investors (Table 6.4), which rules out ordinary noise propagation as the explanation and pins the effect specifically on centrality. This distinction, between a feature being predictively useful and being structurally load-bearing, is what standard accuracy metrics miss.

The methodological contribution here is comparing engineered network features against a heterogeneous GNN under a shared perturbation stress test. This allowed for the results to answer the core question posed at the start: both tabular models and the GNN are broadly robust to random noise but distinctly fragile to targeted attacks on the network. Fragile models might post small ROC-AUC gains, but that's not enough for VC applications, where stability under perturbation matters more than a marginal accuracy increase. That's the distinction this study makes: discriminative performance versus rank stability. A model selection process that compares only clean-graph ROC-AUC would have picked the GNN on the strength of its statistically insignificant point estimate, without ever surfacing the flip-rate and rank-stability problems the perturbation tests found.

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APPENDICES

Appendix A: Entry-window cohort-year and sector distribution (n = 11,797)

Table 8.1: Entry-window cohort-year and sector distribution before final exclusions (n = 11,797)

Cohort year	N	Sector	N	% of sample
2010	1,111	Technology	7,331	62.1%
2011	1,372	Healthcare	1,887	16.0%
2012	1,364	Industrials	871	7.4%
2013	1,475	Consumer Cyclicals	652	5.5%
2014	1,513	Consumer non-cyclicals	456	3.9%
2015	1,682	Financials	211	1.8%
2016	1,585	Basic Materials / Energy / Other	375	3.2%
2017	1,695	Unknown	14	0.1%
Total	11,797	Total	11,797	100.0%

Appendix B: Full statistical significance testing

Table 8.2: Full statistical significance testing, headline network-vs-baseline comparisons

Comparison	AUC A	AUC B	ΔAUC	DeLong Z	DeLong p	Boot CI (low, high)	Sig. (α = .05)	Sig. Bonferroni	Sig. BH-FDR
LogReg_Network vs LogReg_Baseline	0.6525	0.6450	0.0074	2.104	0.0353	(0.0006, 0.0144)	True	False	False
RandomForest_Network vs RandomForest_Baseline	0.6464	0.6489	-0.0025	-0.347	0.7284	(-0.0171, 0.0112)	False	False	False
HistGBM_Network vs HistGBM_Baseline	0.6431	0.6457	-0.0026	-0.366	0.7141	(-0.0167, 0.0112)	False	False	False
XGBoost_Network vs XGBoost_Baseline	0.6428	0.6434	-0.0007	-0.104	0.9169	(-0.0136, 0.0120)	False	False	False
LightGBM_Network vs LightGBM_Baseline	0.6434	0.6408	0.0026	0.357	0.7211	(-0.0122, 0.0169)	False	False	False
GNN vs RandomForest_Baseline	0.6582	0.6489	0.0093	1.102	0.2703	(-0.0073, 0.0258)	False	False	False
GNN vs LogReg_Network	0.6582	0.6525	0.0058	0.961	0.3365	(-0.0062, 0.0173)	False	False	False
LogReg_Network vs RandomForest_Baseline	0.6525	0.6489	0.0035	0.524	0.6004	(-0.0098, 0.0166)	False	False	False

Table 8.3: Bootstrap 95% confidence intervals on clean-graph ROC-AUC, all models

Model	**ROC-AUC**	**95% CI low**	**95% CI high**
LogReg_Baseline	0.6450	0.6215	0.6682
RandomForest_Baseline	0.6489	0.6262	0.6716
HistGBM_Baseline	0.6457	0.6232	0.6678
XGBoost_Baseline	0.6434	0.6208	0.6658
LightGBM_Baseline	0.6408	0.6180	0.6635
LogReg_Network	0.6525	0.6298	0.6752
RandomForest_Network	0.6464	0.6242	0.6684
HistGBM_Network	0.6431	0.6209	0.6650
XGBoost_Network	0.6428	0.6206	0.6646
LightGBM_Network	0.6434	0.6211	0.6651
GNN	0.6582	0.6360	0.6805

Appendix C: Supplementary perturbation checks

The following perturbations supplement the core tests reported in Table 4.2.

Table 8.4: Supplementary perturbation types and mechanisms

Perturbation type	**Model**	**Mechanism**	**Budget (seeds)**
Random edge addition	Both	New edges added uniformly at random between existing investors	0.5–10% of edges (5 seeds)
Edge rewiring	Both	Degree-preserving double-edge swaps	0.5–10% of edges (5 seeds)
Node-feature noise	GNN	Jitter scaled to each feature column's standard deviation	0.5–10% (5 seeds)

Table 8.5: Full perturbation comparison at the 1% and 10% budgets. Each model cell reports ΔROC-AUC, flip rate, and Spearman ρ, in this order

Budget	Perturbation type	RandomForest	HistGBM	LogReg	XGBoost	LightGBM	GNN
1%	Random edge deletion	-0.00038 2.66% 0.9921	0.00096 3.64% 0.9803	-0.00102 1.49% 0.9974	0.00344 2.72% 0.9817	-0.00264 6.94% 0.9627	0.00036 7.51% 0.9899
1%	Preferential edge addition	-0.00026 2.70% 0.9922	0.00094 3.61% 0.9806	-0.00106 1.50% 0.9974	0.00342 2.79% 0.9817	-0.00276 6.91% 0.9629	0.00034 7.53% 0.9901
1%	Degree-aware deletion	-0.00044 2.68% 0.9920	0.00106 3.61% 0.9802	-0.00110 1.50% 0.9973	0.00374 2.77% 0.9816	-0.00246 6.93% 0.9631	0.00038 7.46% 0.9902
1%	Random edge addition	-0.00014 2.82% 0.9916	0.00144 3.79% 0.9797	-0.00096 1.55% 0.9969	0.00298 2.97% 0.9812	-0.00300 6.98% 0.9620	0.00006 8.12% 0.9878
1%	Edge rewiring	-0.00018 0.23% 0.9998	-0.00054 0.21% 0.9989	0.00000 0.07% 1.0000	-0.00020 0.23% 0.9990	0.00004 0.74% 0.9978	0.00020 7.30% 0.9898
1%	Node-feature noise	–	–	–	–	–	0.00000 0.02% 1.0000
10%	Random edge deletion	-0.00026 2.78% 0.9918	0.00126 3.64% 0.9797	-0.00112 1.50% 0.9973	0.00294 2.92% 0.9809	-0.00250 6.97% 0.9619	0.00062 7.07% 0.9895
10%	Preferential edge addition	0.00016 2.79% 0.9918	0.00124 3.58% 0.9807	-0.00096 1.59% 0.9970	0.00308 2.87% 0.9815	-0.00206 6.90% 0.9634	0.00102 7.37% 0.9891
10%	Degree-aware deletion	-0.00006 3.07% 0.9908	0.00064 3.94% 0.9786	-0.00124 1.69% 0.9955	0.00270 3.15% 0.9801	-0.00266 6.99% 0.9602	0.00096 6.84% 0.9911
10%	Random edge addition	-0.00004 3.94% 0.9879	0.00118 4.28% 0.9761	-0.00114 1.94% 0.9949	0.00268 3.69% 0.9765	-0.00450 7.73% 0.9568	-0.00022 9.10% 0.9828
10%	Edge rewiring	-0.00024 1.15% 0.9980	-0.00058 1.30% 0.9939	0.00020 0.23% 0.9999	0.00092 1.39% 0.9943	-0.00028 3.22% 0.9859	0.00048 6.15% 0.9888
10%	Node-feature noise	–	–	–	–	–	-0.00008 0.87% 0.9992